Faculty of Engineering

Module Title: Engineering Chemistry

Revision

A. Property: - any characteristic that can describe or identify matter.

- Properties can be classified as
  - **Intensive properties:** like temperature and melting point have values that do not depend on the amount of sample
  - **Extensive properties:** like length and volume have values that do depend on the amount of sample

B. Pressure: a force (F) exerted per unit area (A), where force is defined as mass (m) times acceleration (a)

$$P = F/A = (m \times a)/A$$

1 atm= 760 mmHg (torr)

$=1.01325*10^5$ Pa

$= 1.01325*10^2$ kPa

= 1.01325 bar

=76 cmHg

C. Volume: The amount of space occupied by an object

$1m^3=1000$ L$=1000$ dm^3

$1dm^3=1$ L

1L=1000 ml (cm^3)

D. Temperature:

$^0K = T\ ^0C + 273.15$

$^0F = 1.8\ T\ ^0C + 32$

E. Amount of substance:

Number of moles = mass/ molecular weight

Kmole=kg/M.Wt

mole= gr/M.Wt

mmloe= mgr/M.Wt

Kmole=10^3 mole, and mole =10^3 mmole

The fundamental and some derived units of measure

Physical Quantity	SI units	Fr units
mass	**Kg**	**gr**
Length	**m**	**cm**
Area	m^2	cm^2
Volume	m^3	cm^3
Time	**sec**	**sec**
Temperature	**K**	**K**
Density	Kg/m^3	gm/cm^3
Speed	m/s	cm/s
Acceleration	m/s^2	cm/s^2
Force	$N\ (kg.m)/s^2$	$gm.cm/s^2$
Pressure	$Pa\ (N/m^2)$	atm
Energy	$J\ (kg.m^2)/s^2$	cal "1 cal=4.18 J"
Amount of substance	**Kmol**	**Mol (gmol)**

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Sheet 1

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1. Typical atmospheric pressure on top of mountain Everest (29,035 ft) is 265 mmHg. Convert this value to pascals, atmospheres, and bars.

   Answer:

   Use the conversion table

2. **Convert the following: -**

   - Pressure of 1150 torr to atm, Pa, cmHg, bar, mmHg
   - Carbon dioxide volume of 500 cm^3 to L, mL, $dm^3$and m^3
   - Temperature of 25 ℃ to K and F

   Answer:

   Use the conversion table

3. Calculate the number of moles of 50 gr of nitrogen gas, 640 mgr of oxygen

   [atomic weight: - N=14 O=16]

   Answer:

   $n_i = m_i/\text{molecular weight}$

   $n_{O2} = 640/(16\times2) = 20$ mmole $=0.02$ mole

   $n_{N2} = 50/(14\times2) = 0.786$ mole

4. How many moles of methane gas (CH_4) are in a storage tank with a volume of 1.0×10^5 L at STP? And how many grams?

   Answer:

   STP T= 0 ℃ +273.15 =273.15 K P = 1atm

   No. Of moles = volume of gas \ volume per mole at STP

   $= 1.0 \times 10^5$ L\ 22.4 L\mole =4464.29 mol

   Mass of methane gas (CH_4)= No. Of moles * Molecular weight

   = 4464.29 mol * (12+4*1)= 71428.64 g

5. An Aerosol spray can with a volume of 350 mL contains 3.2 g of propane gas (C_3H_8) as propellant. What is the pressure in atmospheres of gas in the can at 20 °C?

   Answer:

   T= 20 ℃ +273.15 =293.15 K P = ? atm

   V = 350 mL =0.350 L

   n = 3.2/(3*12+8) = 0.073 mol

   PV=nRT

   $P \times 0.350 = 0.073 \times 0.082 \times 293.15$

   P = 5.01 atm

6. A helium gas cylinder used to fill balloons has a volume of 43.8 L and a pressure of 1.51×10^4 kPa at 25.0 °C. How many moles of helium are in the tank?

Answer:

T= 25 ℃ +273.15 =298.15 K P = 1.51×10^4 kPa = 149.03 atm

V= 43.8 L

PV=nRT

$149.03 \times 43.8 = n \times 0.082 \times 298.15$

n = 266.99 mol

7. What final temperature (°C) is required for the pressure inside an automobile tire to increase from 2.15 atm at 0 °C to 2.37 atm, assuming the volume remains constant?

Answer:

P_1 =2.15 atm T_1= 0 ℃ +273.15 K

P_2= 2.37 atm T_2= ? ℃

$P \alpha T$

$P_1/P_2 = T_1/T_2$

T_2 = 2.37 *273.15 / 2.15 = 301.1 K

T_2 = 27.95 ℃

8. Define:

- **Boyle's law**

  $V \propto 1/P$ or PV = Constant at fixed n and T

- **Charles's law**

  $V \propto T$ or V/T=Constant at fixed n and P

- **Avogadro's law**

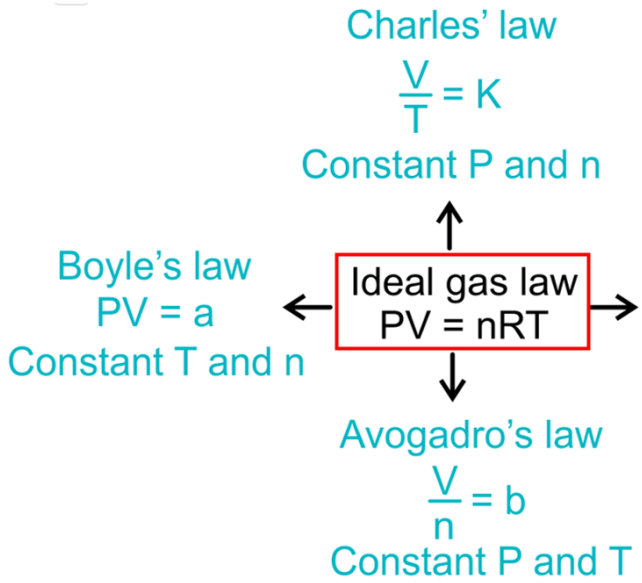
  $V \propto n$ or V/n=Constant at fixed T and P

- **Mole fraction**

  The ratio of the number of moles of that component to the total number of moles of all components in the mixture.

  $$x_i = n_i / n_t$$

- **Ideal gas law**



Where:

- P is the pressure of the gas,
- V is the volume of the gas,
- n is the amount of substance of gas (also known as number of moles),
- R is the ideal, or universal, gas constant
- T is the absolute temperature of the gas

9. Choose the correct answer:

A. The Correct statement is:

- **<u>Gases at low pressure or high temperature behave as an ideal gas</u>**
- Gases at high pressure or low temperature behave as an ideal gas
- Gases at high density behave as an ideal gas
- None of the mentioned

B. What is the density of Carbon dioxide (CO_2) in kg/m^3 at 27 °C and 100 kPa?

- **<u>1.76</u>**
- 2.76
- 3.76
- 4.76

P × molecular weight = density ×R×T

100 × 1000 × (12+16×2) = Density ×8.314×10^3 J/kmol.K × (27+273)

Density = (10^5 × 44)/ (8.314×10^3×300) =1.764 kg/m^3

C. Atmospheric pressure and 0°C conditions are known as:

- Room temperature and pressure
- **<u>Standard temperature and pressure</u>**
- Atmospheric temperature and pressure
- None of the mentioned

D. For ideal gas, which pair of variables is inversely proportional to each other (if all other conditions remain constant)?

- P, T
- **P, V**
- V, T
- n, V

10. Uranium hexafluoride is a molecular solid used to purify the uranium isotope needed to fuel nuclear power plants. Assume that you have a 22.9 L vessel that contains 512.9 g of UF6 at 70.0 °C. What is the pressure in the vessel calculated using the ideal gas law and the van der Waals equation? (For UF6, a= 15.8 L^2.atm/mol^2, b= 0.1128 L/mol). (Molar mass of UF6 is 352.02 g/mol)

Answer:

V= 22.9 L m = 512.9 g T= 70 + 273= 343 K

$$n = \frac{m}{Molar\ Mass} = \frac{512.9\ g}{352.02\ g/mol} = 1.46\ mol$$

For ideal gas

PV=nRT

$$P = \frac{n \times R \times T}{V} = \frac{1.46\ mol \times 0.082\ \frac{atm.L}{mol.K} \times 343\ K}{22.9\ L} \approx 1.8\ atm$$

For Real gas

$[P + (an^2/V^2)]\ [V - nb] = nRT$

$$\mathrm{P} = \frac{nRT}{V - nb} - \frac{an^2}{V^2}$$

$$P = \frac{1.46\ mol \times 0.082 \frac{atm.L}{mol.K} \times 343\ K}{22.9\ L - 1.46\ mol \times 0.1128\ L/mol} - \frac{15.8\ \frac{L^2.atm}{mol^2} \times 1.46^2 mol^2}{22.9^2\ L^2}$$
$$= 1.81 - 0.064 = 1.75\ atm$$